

LSAR: Sparse Lexical Representation Learning for Efficient and Interpretable Audio Retrieval

Map audio directly to inspectable lexical evidence - then retrieve with a standard inverted index.

10M-ITEM STRESS TEST

0.97 ms

CPU query latency with a 44 MB sparse index
445× less memory than the CLAP dense bank

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 github.com/Ameeeeeeeeeeeeeee/LSAR

Introduction

Real-time audio retrieval for multimodal RAG

Audio is becoming a primary query and knowledge modality for MLLMs, but today's retrievers force a poor trade-off. ASR cascades offer lexical matching yet lose non-verbal evidence and propagate transcription errors. Dense models preserve semantics but require opaque, memory-heavy vector search. **LSAR introduces a third path:** learn a non-negative, highly sparse audio representation in a text vocabulary space, expose activated terms as evidence, and retrieve candidates using dynamic-pruned posting lists.

Motivation: the missing middle

ASR → text cascade

- Non-verbal events disappear
- Errors propagate into retrieval
- Generation adds indexing latency

Dense audio retrieval

- Relevance is difficult to inspect
- Large embedding banks
- Exact search scales with corpus size

Design target: Speech + sound coverage · lexical evidence · compact posting-list search

Core contributions

1 New retrieval primitive

Direct audio-to-lexical mapping compatible with existing sparse search infrastructure.

2 Cross-modal alignment

Learn audio term activations from a frozen SPLADE-v3 text teacher during training.

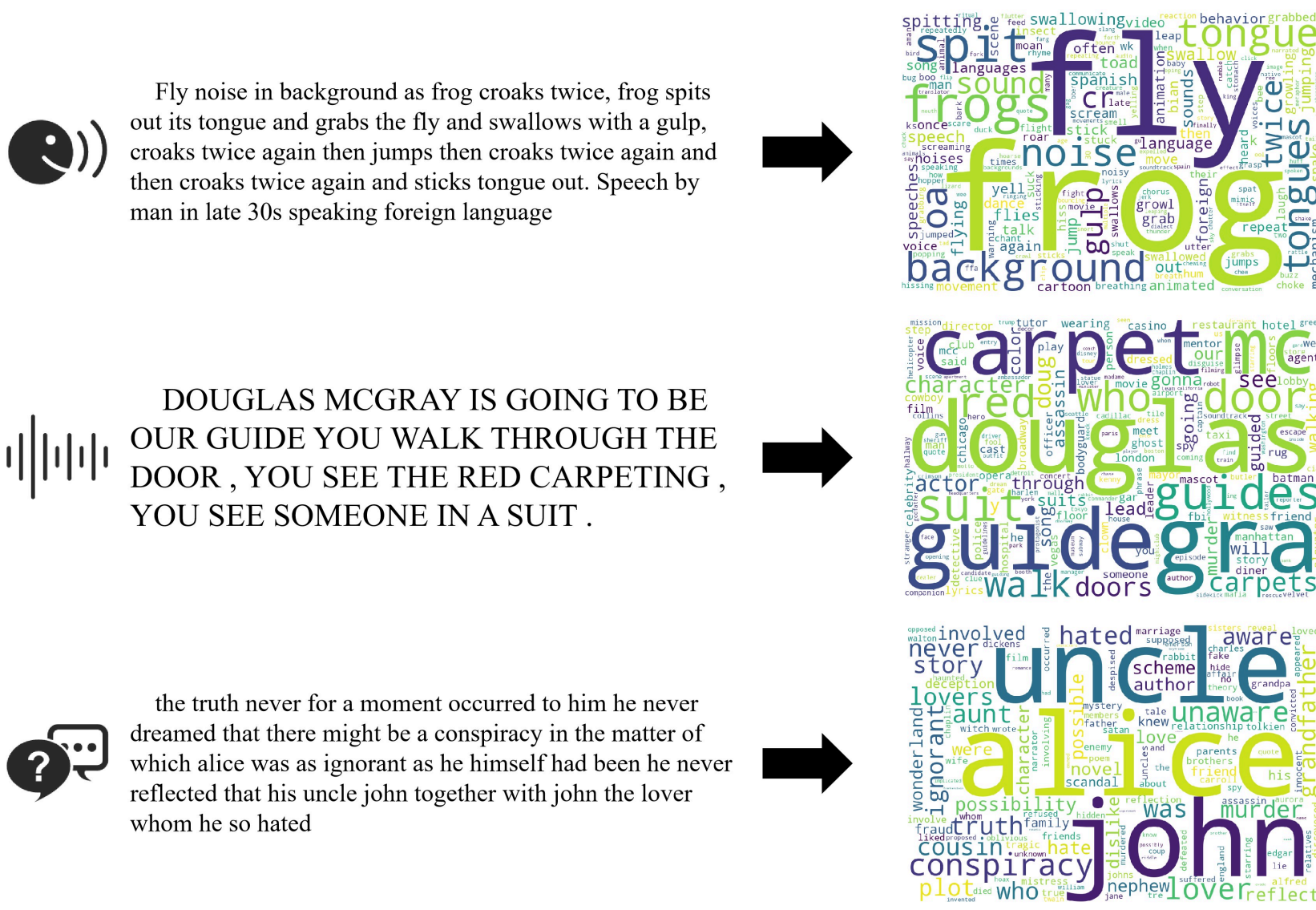
3 Three complementary views

Content, Caption, and Logic heads ground words, events, and QA associations.

4 Practical first stage

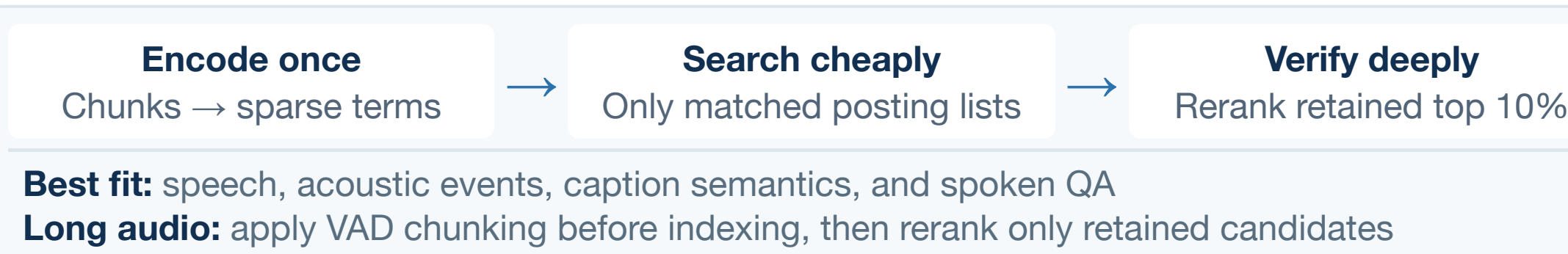
Prune aggressively at high recall, expose keyword evidence, and plug directly into downstream dense rerankers.

Inspectable lexical evidence

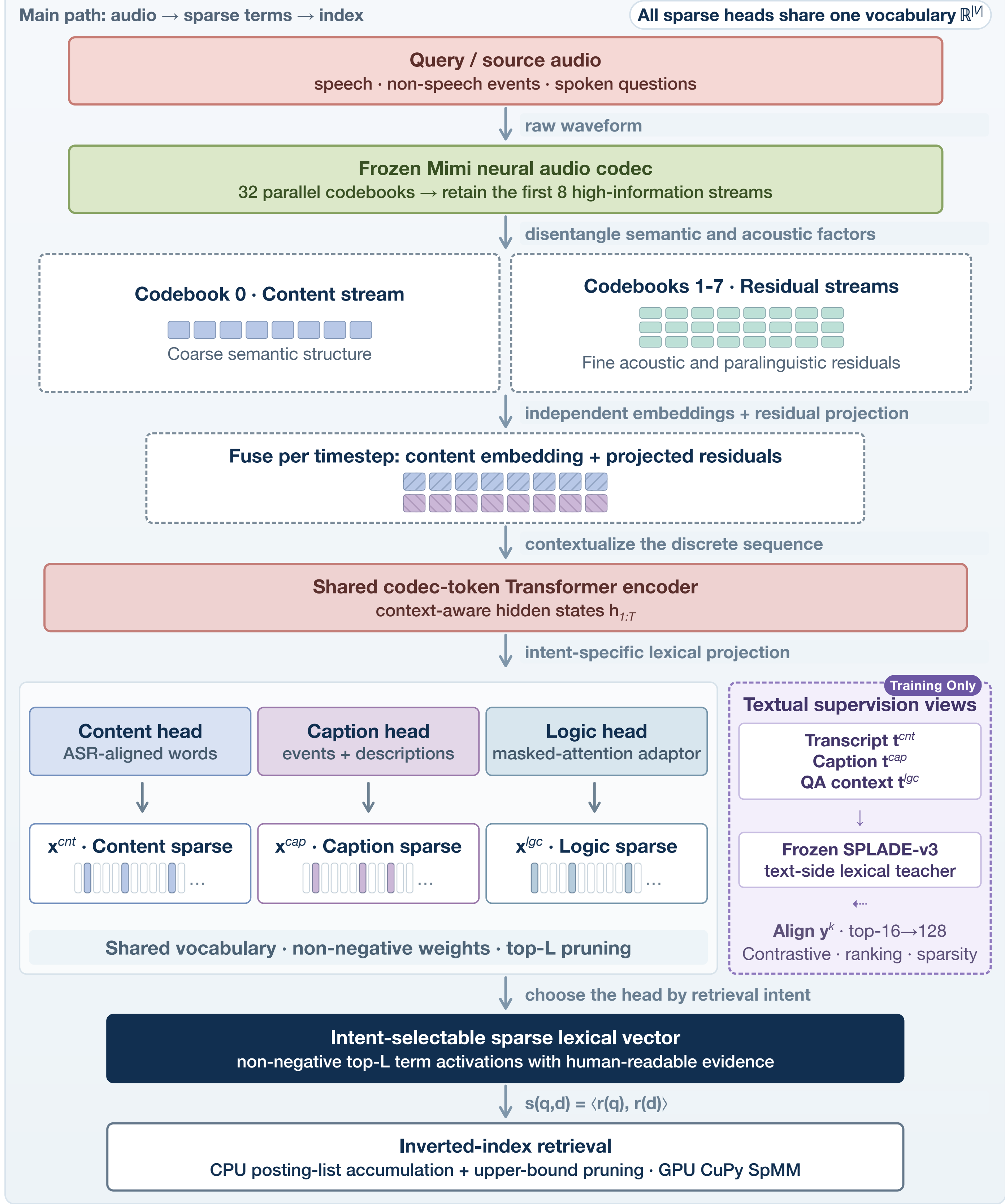


Why was a clip retrieved? Each head exposes non-negative vocabulary activations. Content highlights spoken words. Caption surfaces events and background sounds. Logic emphasizes context-dependent associations. The terms can be audited, passed to a reranker, or shown as a compact provenance trail.

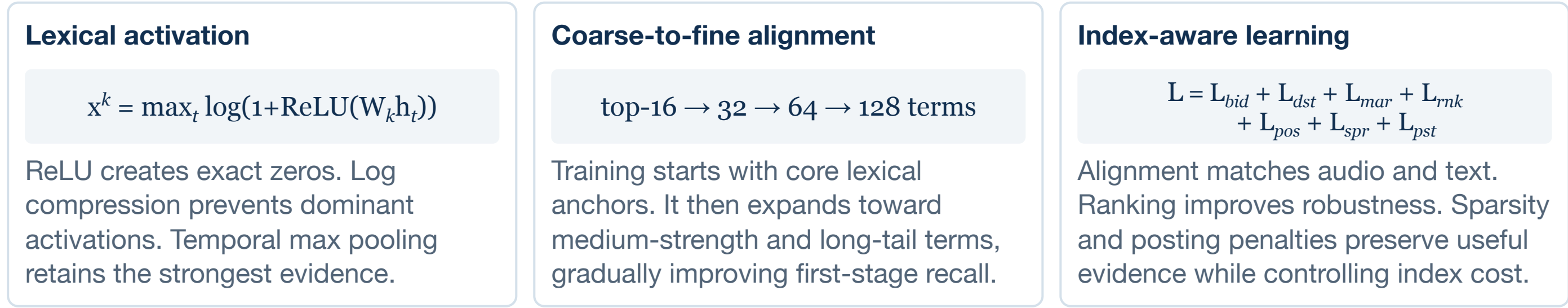
Retrieval workflow



Architecture: inference and training



Three representation mechanics



Training data & evaluation

Content LibriSpeech + GigaSpeech-L Transcript-level grounding 3,460 h	Caption AudioCaps + TextrolSpeech + SpeechCraft descriptions ≈760k clips	Logic LibriSQA + SLUE-SQA-5 + Spoken SQuAD ≈200k QA	Evaluation 4 in-domain + 2 zero-shot ESC-50 and Clotho transfer 6 benchmarks
Training: 8× A100 80GB · AdamW · batch 128 · LR 5×10 ⁻⁵ · Caption weight 1.5-2.0. Inference: frozen Mimi with direct sparse lexical projection and no ASR.			

Headline results

1.00 R@10% on LibriSpeech and AudioCaps	7.11% minimum AudioCaps subset for full sparse recall	90% of candidates removed before dense reranking
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First-stage retrieval

Dataset	Setting	Recall@10%	Min. Sub.	Latency
LibriSpeech	In-domain	1.0000	8.96%	3.09 ms
AudioCaps	In-domain	1.0000	7.11%	3.43 ms
ESC-50	Zero-shot	0.9570	14.00%	2.17 ms
Clotho	Zero-shot	0.8173	17.92%	3.58 ms
LibriSQA	In-domain	0.8352	16.73%	2.96 ms
SLUE-SQA	In-domain	0.7449	24.94%	3.02 ms

Recall@10%: fraction of queries whose target appears within the top-ranked 10% of candidates.
Min. Sub.: smallest retained corpus fraction that still preserves full sparse-retrieval recall.
Zero-shot: evaluation without task-specific training or benchmark-specific fine-tuning.

10M-scale efficiency

Index memory · 10M items LSAR CPU CLAP dense bank 44 MB 19,613 MB 445× less index memory	Single-query latency LSAR CPU CLAP-ANN GPU CLAP-Exact GPU 0.97 ms 9.88 ms 50.9 ms
CPU index: posting-list accumulation with upper-bound pruning.	Complexity: proportional to matched posting lists, not all N dense vectors.

Two-stage retrieval

CLAP direct · AudioCaps R@1 43.1% rerank all candidates	→	LSAR → CLAP · AudioCaps R@1 52.1% rerank top 10% only
+9.0 points R@1 and 93.5% R@10. LSAR is not meant to replace the final ranker: it cheaply removes irrelevant items and can improve the candidate distribution seen by CLAP, a cross-encoder, or an LLM verifier.		

Why three heads?

Speech +8.3 R@10% points on LibriSpeech: Content-only 0.9167 → Full 1.0000.	Environmental sound +19.7 points on ESC-50: Caption-only 0.7600 → Full 0.9570.
Caption retrieval +16.4 points on Clotho: Caption-only 0.6532 → Full 0.8173.	Spoken QA +41.0 points on LibriSQA: Logic-only 0.4253 → Full 0.8352.

Deployment scope

Scope and trade-off. LSAR optimizes high-recall candidate generation, not standalone top-1 dominance. Very subtle paralinguistic signals may remain better represented by dense models, and multilingual retrieval requires a multilingual sparse teacher.		
ASR-free serving Audio maps directly to postings	Two backends CPU postings or GPU SpMM	Auditable Top-L terms explain candidates